第六次作业

习题五 1(4)(5) 2(2)(7)(8) 3(2)(8)(9)

5 8(3) 9(2) 11(1)

1(4) $\cos\frac{1}{z}=1-\frac{1}{2!}(\frac{1}{z})^2+\frac{1}{4!}(\frac{1}{z})^4+\cdots$

故 $z\cos\frac{1}{z}=z-\frac{1}{2!}\frac{1}{z}+\frac{1}{4!}(\frac{1}{z})^3+\cdots$

$z=0$ 为 $z\cos\frac{1}{z}$ 的本性奇点

(5) $\frac{z^2}{\cosh z-1}=\frac{z^2}{\frac{z^2}{2}+\frac{z^4}{4!}+\cdots}$

$z=0$ 是 z^2 的二级零点，是 $\cosh z-1$ 的二级零点，

故 $z=0$ 是 $\frac{z^2}{\cosh z-1}$ 的可去奇点

2. (2) $\tan z=\frac{\sin z}{\cos z}$

$\cos z=0 \Rightarrow z=\frac{\pi}{2}+k\pi$，为孤立单极点

故 $\operatorname{Res}[\tan z, z_R]=-1$

(7) $z=-1$ 为 $\sin\frac{z}{z+1}=\sin(1-\frac{1}{z+1})=\sin 1\cos\frac{1}{z+1}-\cos 1\sin\frac{1}{z+1}$

$=\sin 1-\cos 1\frac{1}{z+1}-\sin 1\frac{1}{2!}(\frac{1}{z+1})^2+\cos 1\frac{1}{3!}(\frac{1}{z+1})^3-\cdots$

故 $\operatorname{Res}[\sin\frac{z}{z+1}, -1]=-\cos 1$

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(8) $z=1$，设 $z-1=t$，则 $z^2e^{\frac{1}{z-1}}=(t+1)^2e^{\frac{1}{t}}$

$=(t+1)^2+\frac{(t+1)^2}{t}+\frac{(t+1)^2}{2t^2}+\frac{(t+1)^2}{6t^3}+\cdots$，$\frac{1}{t}$ 系数为 $\frac{13}{6}$

故 $\operatorname{Res}[z^2e^{\frac{1}{z-1}},1]=\frac{13}{6}$

3. (2) $\oint_C \frac{z}{(z-1)(z-2)^2}dz$ $C:|z-2|=\frac{1}{2}$

$z=2$，为 $\frac{z}{(z-1)(z-2)^2}$ 二级极点

原式 $=2\pi i \operatorname{Res}[\frac{z}{(z-1)(z-2)^2},2]$

$=2\pi i \lim_{z\to 2}\frac{-1}{(z-1)^2}=-2\pi i$

(8) $\oint_{|z|=1}\frac{\cos(e^{-z})}{z^2}dz$

$z=0$，为 $\frac{\cos(e^{-z})}{z^2}$ 二级极点

原式 $=2\pi i \operatorname{Res}[\frac{\cos(e^{-z})}{z^2},0]$

$=2\pi i \; e^{-z}\sin(e^{-z})\big|_{z=0}=2\pi i\sin 1$

(9) $z_k=k\pi$，$k=-2,-1,0,1,2$

原式 $=2\pi i\sum \operatorname{Res}[\cot z;z_k]=10\pi i$

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$$\text{设 } f_1(z)=\frac{\varphi_1(z)}{z-z_0},\ f_2(z)=\frac{\varphi_2(z)}{z-z_0}$$

5. z_0是$f_1(z)\cdot f_2(z)$的单极点 $\Leftrightarrow$ z_0是$\frac{1}{f_1(z)}$,$\frac{1}{f_2(z)}$的单零点

故z_0是$\frac{1}{f_1(z)+f_2(z)}$的二级零点 $\Leftrightarrow$ z_0是$f_1(z)+f_2(z)$的二级极点

$$\begin{aligned}\operatorname{Res}[f_1(z)\cdot f_2(z);z_0]&=\lim_{z\to z_0}\left[(z-z_0)^2\frac{\varphi_1(z)}{z-z_0}\frac{\varphi_2(z)}{z-z_0}\right]'\\&=\varphi_1'(z_0)\varphi_2(z_0)+\varphi_1(z_0)\varphi_2'(z_0)\\&=\varphi_1'(z_0)\operatorname{Res}[f_2(z);z_0]+\varphi_2'(z_0)\operatorname{Res}[f_1(z);z_0]\end{aligned}$$

8(3). 设$z=e^{i\theta}$ 则$dz=ie^{i\theta}d\theta$

$$d\theta=\frac{1}{ie^{i\theta}}dz$$

$$\text{原式}=\oint_{|z|=1}\frac{1}{1-2\rho\frac{z+\frac{1}{z}}{2}+\rho^2}\frac{1}{iz}dz$$

$$=\oint_{|z|=1}\frac{-i}{(-\rho z+1)(z-\rho)}dz$$

$$=2\pi i\operatorname{Res}\left[\frac{-i}{(-\rho z+1)(z-\rho)};\rho\right]$$

$$=\frac{2\pi}{1-\rho^2}$$

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10(2) $\int_0^{\infty} \frac{x^2}{(x^2+a^2)^2}\,dx$

$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{x^2}{(x^2+a^2)^2}\,dx$

$= \frac{1}{2} 2\pi i \cdot \operatorname{Res}\left[\frac{x^2}{(x^2+a^2)^2}, ia\right]$

$= \pi i \lim_{x\to ia}\left[\frac{x^2}{(x+ia)^2}\right]' =$

$= \pi i \lim_{x\to ia} \frac{2xia}{(x+ia)^3} = -\frac{\pi}{4a}$

11(1) $\int_0^{\infty} \frac{x\sin x}{1+x^2}\,dx$

$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{x\sin x}{1+x^2}\,dx$

$= \operatorname{Im}\left(\frac{1}{2}\int_{-\infty}^{\infty} \frac{xe^{ix}}{1+x^2}\,dx\right)$

$= \operatorname{Im}\left\{\pi i \operatorname{Res}\left[\frac{xe^{ix}}{1+x^2}, i\right]\right\}$

$= \operatorname{Im}\left(\frac{\pi i e^{-1}}{2}\right)$

$= \frac{\pi}{2e}$

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